

1. It comprises of nodes which are dependent and independent variables
2. Also the operators.
3. The arcs / edges of this graph denotes the functional process flow.

Let's consider an example:

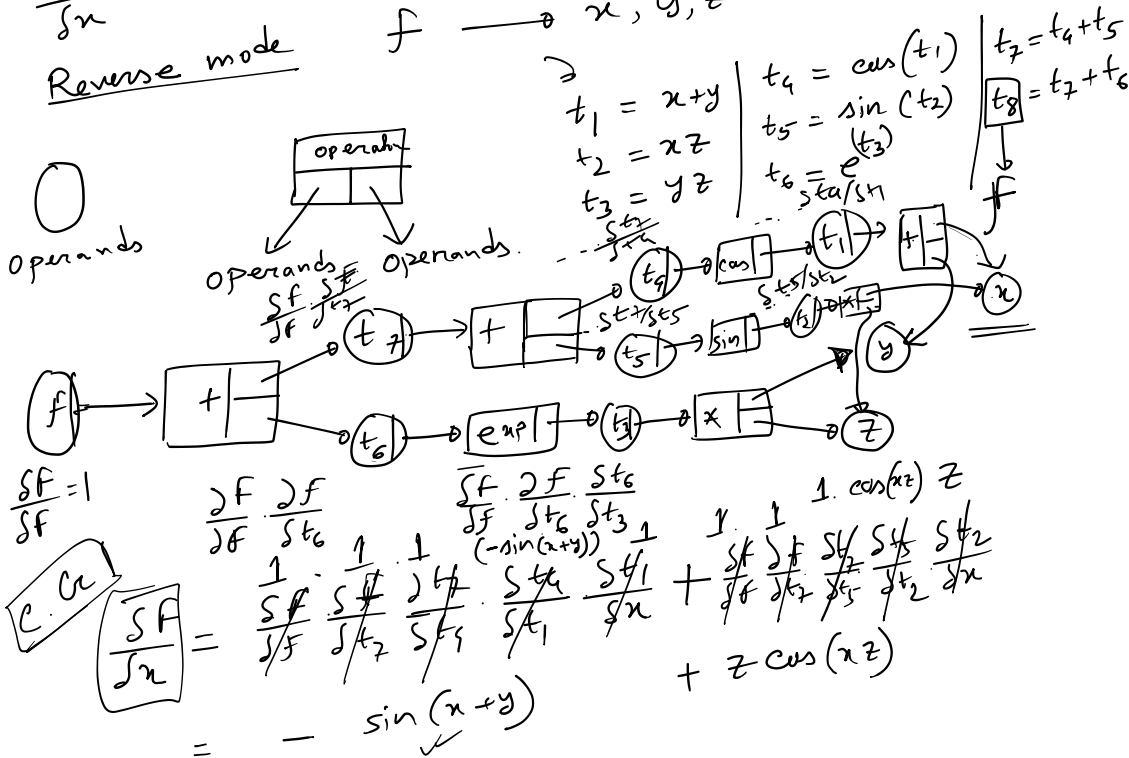
$$\frac{\partial f}{\partial x} = \frac{-\sin(x+y) + z \cos(xz)}{+0}$$

$f(x, y, z) = \cos(x+y) + \sin(xz) + e^{yz}$
Gradient of f with respect to any independent variable like (x, y, z)

$$\frac{\partial f}{\partial x} = ?$$

How C.G. helps us to find such gradients.

Reverse mode $f \rightarrow x, y, z$



$$f(x) = \sin^7 x$$

$$\frac{\partial f}{\partial x} = 7 \sin^6 x \cdot \cos x$$

$$t_1 =$$

$$t_2 =$$

$$t_3 =$$

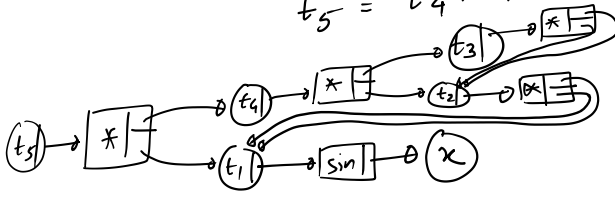
$$t_4 =$$

$$t_5 =$$

$$t_6 =$$

$$f$$

$$t_1 = \sin x \quad \begin{cases} t_2 = t_1 \times t_1 // \sin^2 x \\ t_3 = t_2 \times t_2 // \sin^4 x \\ t_4 = t_3 \times t_2 // \sin^6 x \\ t_5 = t_4 \times t_1 // \sin^7 x \end{cases}$$



incoming count

$$t_1 \rightarrow 3 \rightarrow 2$$

$$t_2 \rightarrow 3 \rightarrow 2$$

$$\left(\frac{\partial f}{\partial t_5} \frac{\partial t_5}{\partial t_4} + \frac{\partial f}{\partial t_4} \frac{\partial t_4}{\partial t_3} + \frac{\partial f}{\partial t_3} \frac{\partial t_3}{\partial t_2} + \frac{\partial f}{\partial t_2} \frac{\partial t_2}{\partial t_1} \right) \frac{\partial t_1}{\partial x}$$

$$= (\sin^6 x + (t_1^2 t_2^2 + \sin^5 x) 2 \sin x) \cos x$$